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Outlook OL	Temperature T	Humidity H	windy W	Rain R
Sunny	Hot	High	weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	weak	yes.
Rainy	Mild	High	weak	yes
Rainy	cool	Normal	weak	yes
Rainy	cool	Normal	Strong	No
Cloudy	Mild	Normal	Strong	yes.
Sunny	cool	Normal	weak	No

Step 1: calculate entropy

$$-\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$-0.5\,(\log_2 0.5) - 0.5\,(\log_2 0.5)$$

$$-0.5\,(-1) - 0.5\,(-1)$$

$$0.5 + 0.5 = \boxed{1}$$

out look (OL)

Sunny: $-\frac{0}{3}\log_2\left(\frac{0}{3}\right) - \frac{3}{3}\log_2\frac{3}{3} \Rightarrow 0$

Cloudy: $-\frac{2}{2}\log_2\left(\frac{2}{2}\right) - \frac{0}{2}\log_2\frac{0}{2} \Rightarrow 0$

Rainy: $-\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right)$

$= -0.66\log_2(0.66) - 0.33\log_2(0.33)$

$= -0.66 \times -0.599 - 0.33(-1.599)$

$= -0.3894 - 0.527$

$= 0.916$

Weighted Entropy:

$\frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.916)$

$0 + 0 + 0.3435$

$= 0.3435$

Info gain:

$1 - 0.344 = \boxed{0.656}$

2) Temperature:

Hot: $-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$

$= -0.5\log_2(0.5) - 0.5\log_2(0.5)$

$-0.5(-1) - 0.5(-1)$

$0.5 + 0.5 = 1$

Mild :-

$$-\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$-0.66\log_2(0.66) - 0.33\log_2(0.33)$$

$$-0.66 \times -0.599 - 0.33 \times -1.599$$

$$0.395 + 0.527$$

$$= 0.922$$

Cool : $-\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$

$$=) -0.33\log_2(0.33) - 0.66\log_2(0.66)$$

$$-0.33 \times -1.599 \quad - \quad 0.66 \times -0.599$$

$$= \quad 0.527 + 0.395$$

$$= 0.922$$

$$\frac{2}{8}(1) + \frac{3}{8}(0.922) + \frac{3}{8}(0.922)$$

$$0.25 + 0.345 + 0.345 = 0.940$$

Info gain :

$$1 - \quad 0.940 = 0.06$$

Humidity:-

High:- $-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$

$-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$

$-0.5\log_2(0.5) - 0.5\log_2(0.5)$

$0.5(-1) - 0.5(-1)$

$= 1$

Normal:- $-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$

$-0.5\log_2(0.5) - 0.5\log_2(0.5)$

$-0.5(-1) - 0.5(-1)$

$0.5 + 0.5 = 1$

weighted entropy:-

$= \frac{4}{8}(1) + \frac{4}{8}(1)$

$0.5 + 0.5 = 1$

Info gain $= 1 - 1 =$ **0**